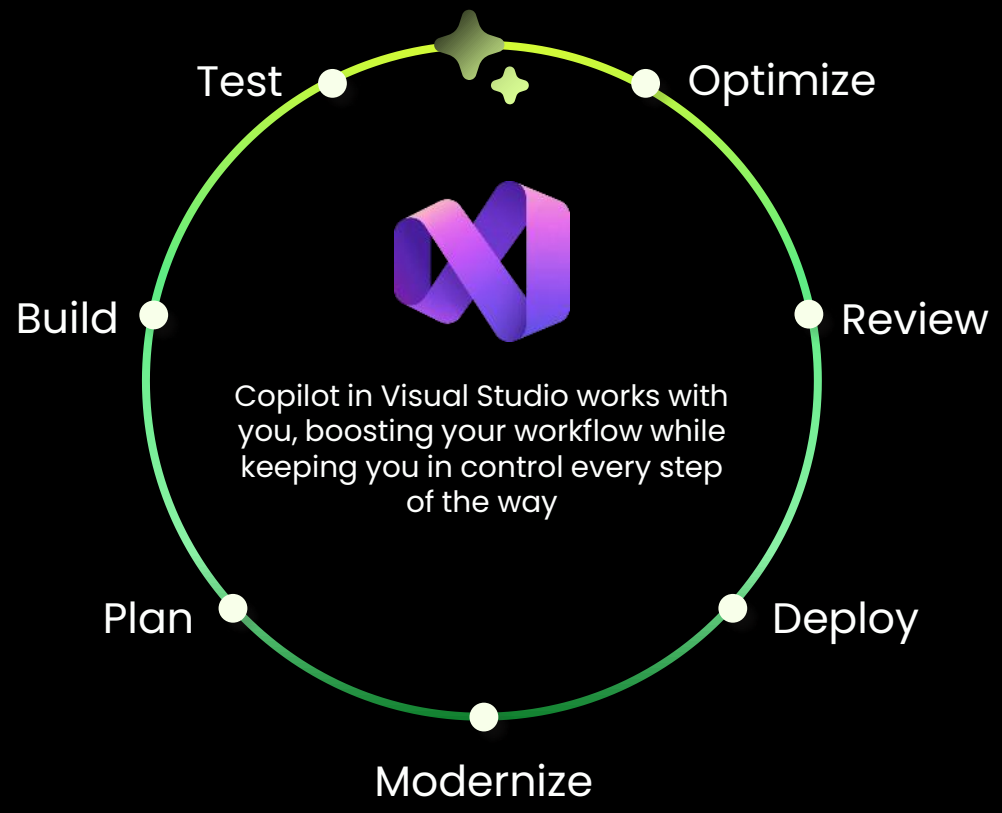
GitHub Copilot Dev Days

Visual Studio + GitHub Copilot

GitHub Copilot Dev Days



Visual Studio 2026

Capabilities

- Debugger
- Profiler
- Test
- Modernize
- GitHub

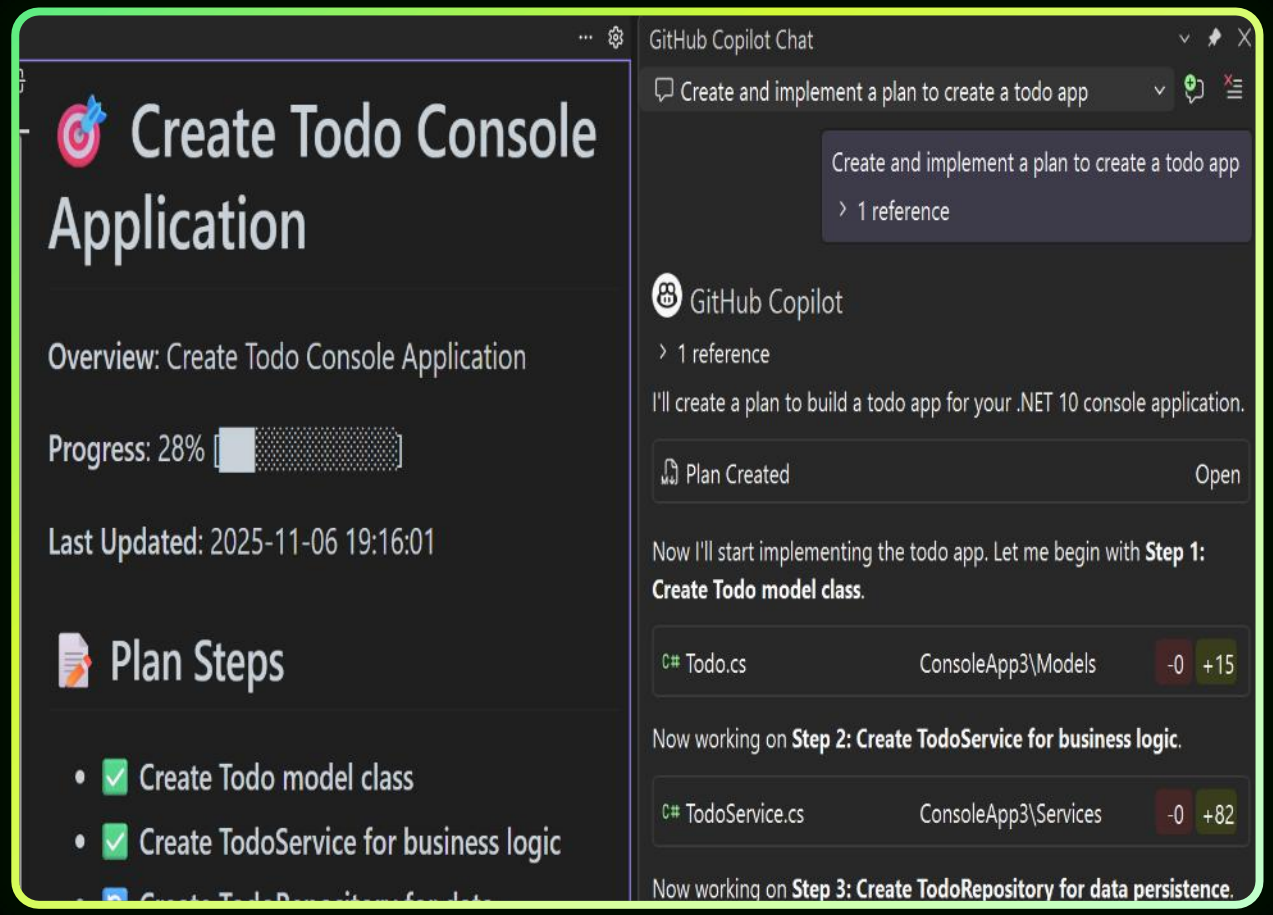
Tools

- Planning
- Microsoft Learn
- NuGet
- GitHub
- Azure MCP server

& more

Plan

Plan


Create Todo Console Application
Overview: Create Todo Console Application
Progress: 28%
Last Updated: 2025-11-06 19:16:01
Plan Steps
Create Todo model class
Create TodoService for business logic
GitHub Copilot Chat
Create and implement a plan to create a todo app
Create and implement a plan to create a todo app
1 reference
GitHub Copilot
1 reference
I'll create a plan to build a todo app for your .NET 10 console application.
Plan Created
Open
Now I'll start implementing the todo app. Let me begin with Step 1: Create Todo model class.
Todo.cs
ConsoleApp3\Models
-0
+15
Now working on Step 2: Create TodoService for business logic.
TodoService.cs
ConsoleApp3\Services
-0
+82
Now working on Step 3: Create TodoRepository for data persistence.

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search TimeZoneHelperWebApp

Debug Any CPU TimeZoneHelperWebApp.Web https INSIDERS

Solution Explorer

Search Solution Explorer (Ctrl+;)

- Solution 'TimeZoneHelperWebApp' (4 of 4 projects)
  - TimeZoneHelperWebApp.ApiService
    - Connected Services
    - Dependencies
    - Properties
    - appsettings.json
    - Program.cs
    - TimeZoneHelperWebApp.ApiService.http
  - TimeZoneHelperWebApp.AppHost
    - Connected Services
    - Dependencies
    - Properties
    - AppHost.cs
    - appsettings.json
  - TimeZoneHelperWebApp.ServiceDefaults
    - Dependencies
    - Extensions.cs
  - **TimeZoneHelperWebApp.Web**
    - Connected Services
    - Dependencies
    - Properties
    - wwwroot
    - Components
    - appsettings.json
    - Program.cs
    - WeatherApiClient.cs

Solution Explorer Git Changes

GitHub Copilot Chat

New Thread

Copilot Chat

I might make mistakes, so check for accuracy.

Suggest how I can optimize my code

Write unit tests for all the methods in #file

How can I improve #method? Suggest code for any improvements

Use # for references, / for commands, and @ for extensions.

/help what can you do?

+ Reference

Ask Copilot or use @workspace

Agent Claude Sonnet 4.5

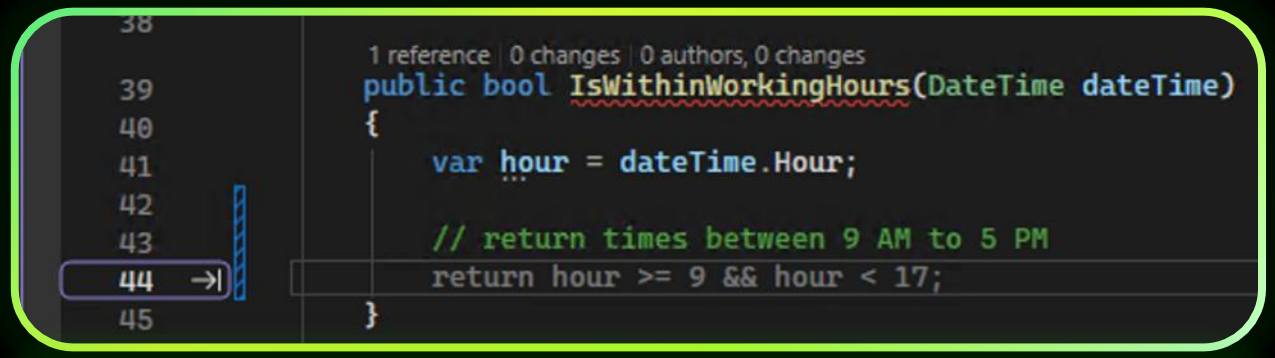
Diagnostic Tools

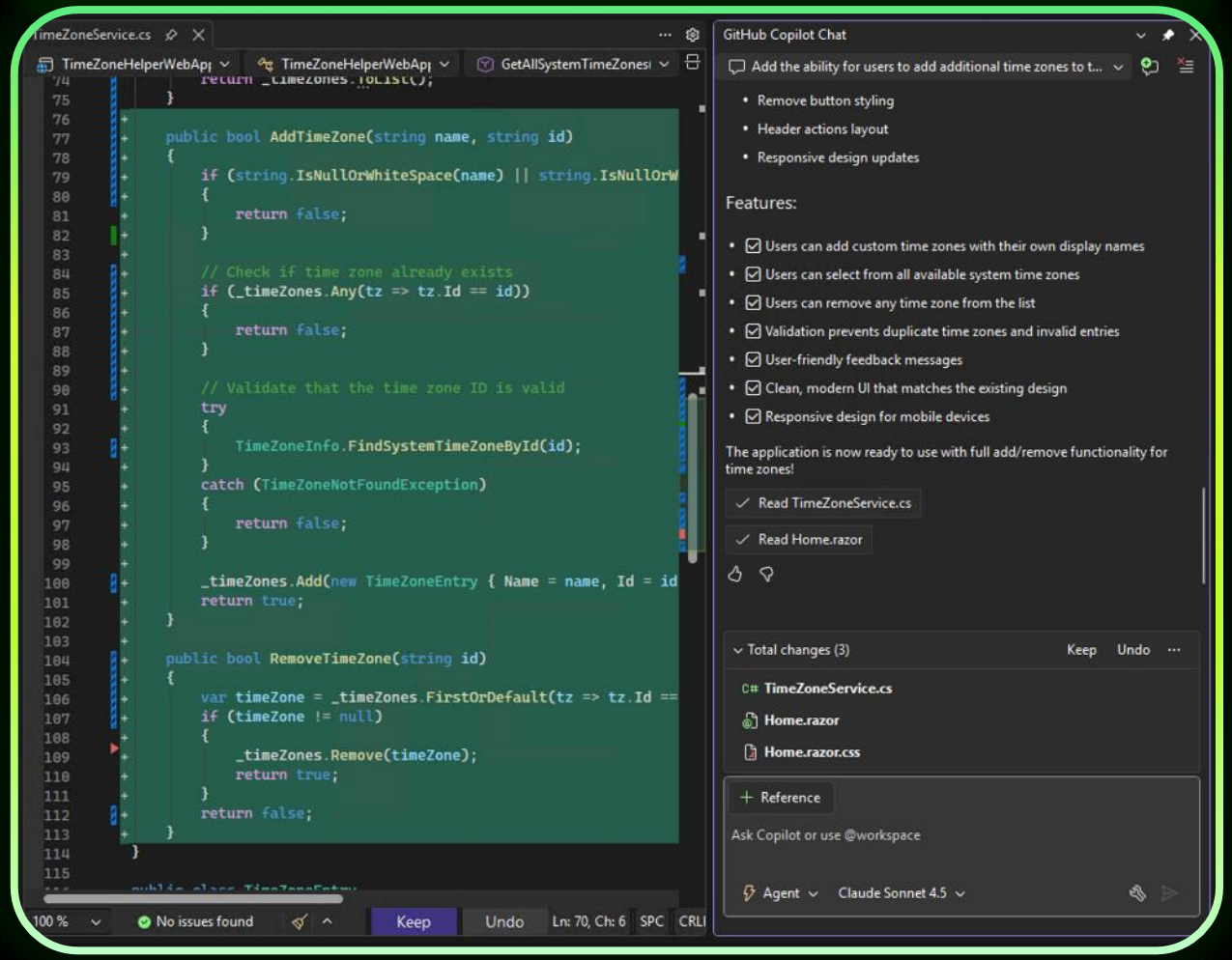
Ready Add to Source Control Select Repository

54°F Cloudy Search

Build

Build


1 reference | 0 changes | 0 authors, 0 changes
public bool IsWithinWorkingHours(DateTime dateTime)
{
var hour = dateTime.Hour;
// return times between 9 AM to 5 PM
return hour >= 9 && hour < 17;
}


TimeZoneService.cs
public bool AddTimeZone(string name, string id)
{
if (string.IsNullOrWhiteSpace(name) || string.IsNullOrW
{
return false;
}
// Check if time zone already exists
if (_timeZones.Any(tz => tz.Id == id))
{
return false;
}
// Validate that the time zone ID is valid
try
{
TimeZoneInfo.FindSystemTimeZoneById(id);
}
catch (TimeZoneNotFoundException)
{
return false;
}
_timeZones.Add(new TimeZoneEntry { Name = name, Id = id
return true;
}
public bool RemoveTimeZone(string id)
{
var timeZone = _timeZones.FirstOrDefault(tz => tz.Id ==
if (timeZone != null)
{
_timeZones.Remove(timeZone);
return true;
}
return false;
}
}
No issues found
Keep
Undo
Ln: 70, Ch: 6
GitHub Copilot Chat
Add the ability for users to add additional time zones to t...
Remove button styling
Header actions layout
Responsive design updates
Features:
Users can add custom time zones with their own display names
Users can select from all available system time zones
Users can remove any time zone from the list
Validation prevents duplicate time zones and invalid entries
User-friendly feedback messages
Clean, modern UI that matches the existing design
Responsive design for mobile devices
The application is now ready to use with full add/remove functionality for time zones!
Read TimeZoneService.cs
Read Home.razor
Total changes (3)
Keep
Undo
TimeZoneService.cs
Home.razor
Home.razor.css
Reference
Ask Copilot or use @workspace
Agent
Claude Sonnet 4.5

Test

GitHub Copilot Chat

@test #C:\Users\cathys\source\repos\eShop-tes...

Test (Preview)

0 references

Summary of Test Generation

Status: Completed Successfully

Total Unit Tests Generated: **59** tests

Files Created/Modified: **7** files

Test Projects Touched: **1** project

- Projects: Basket.UnitTests.csproj

Test Coverage Impact

Initial Coverage: **4.2%**

Final Coverage: **46.6%**

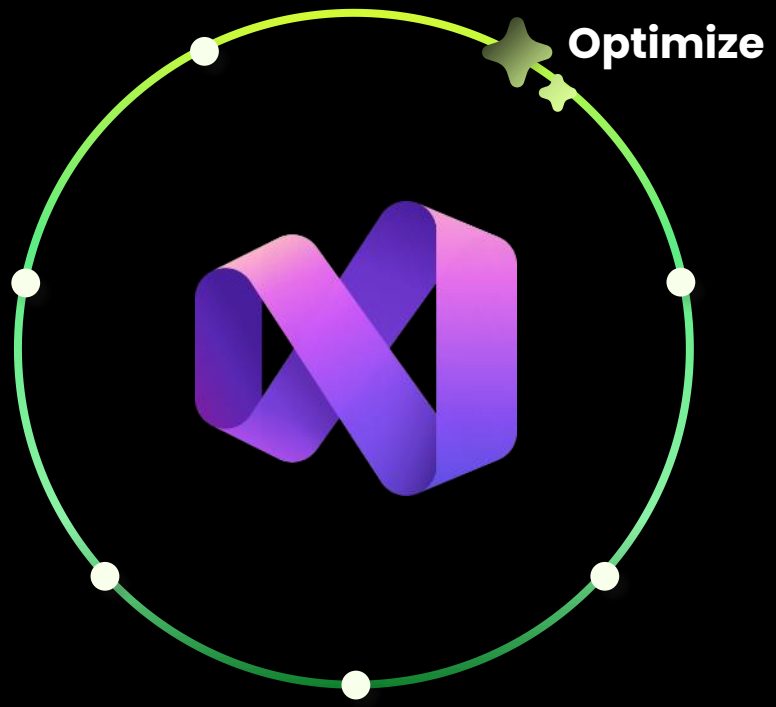
Coverage Change: +**42.3**pp

Quality Indicators

Pass/Fail Summary: All tests build and pass

Insights & Recommendations

Optimize



GitHub Copilot Chat

Analyze the value of the expression 'choice'

Using #Debugger, explain the value of the expression **choice**

Debugger

2 references

I'll help you analyze the value of the **choice** variable. Let me start by understanding the context of your program.

Retrieved 'ConsoleApp3.UI.ConsoleUI'

Now let me check the switch statement in the Main method to see how this choice is being handled:

Read Program.cs

Perfect! Now I have all the context I need to analyze the **choice** variable.

Analysis

Value: "1" (string)

What it represents: The user's menu selection indicating they want to **Add a new Todo** item to the application.

Why you're getting this value:

The program displayed a menu with 8 options via **ui.DisplayMenu()**. The user typed "1" at the console prompt and pressed Enter. **Console.ReadLine()** captured this input as the string **"1"**. The debugger has paused at the switch

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search TimeZoneHelperWebApp

Debug Any CPU TimeZoneHelperWebApp.Web Continue Search Telemetry INSIDERS

Process: [40208] TimeZoneHelp Lifecycle Events Thread: Stack Frame:

TimeZoneService.cs

TimeZoneHelperWebApp.Web | TimeZoneHelperWebApp.Web.Services.TimeZoneService | GetTimeZoneDisplayInfo(DateTime localDateTime, string localT

```
        // Convert local time to UTC first
        var utcTime = TimeZoneInfo.ConvertTimeToUtc(localDateTime, localTimeZone);

        var displayInfoList = new List<TimeZoneDisplayInfo>();

        foreach (var tz in _timeZones)
        {
            var tzInfo = TimeZoneInfo.FindSystemTimeZoneById(tz.Id);
            var convertedTime = TimeZoneInfo.ConvertTimeFromUtc(utcTime, tzInfo);

            displayInfoList.Add(new TimeZoneDisplayInfo
            {
                DisplayName = tz.Name,
                TimeZoneId = tz.Id,
                LocalTime = convertedTime,
                IsWorkingHours = IsWithinWorkingHours(convertedTime)
            });
        }

        return displayInfoList.ToArray();
    }

    1 reference | 0 changes | 0 authors, 0 changes
    public bool IsWithinWorkingHours(DateTime dateTime)
    {
        var hour = dateTime.Hour;
        return hour >= 9 && hour < 17; // 9am to 5pm (17:00)
    }

    0 references | 0 changes | 0 authors, 0 changes
    public string[] GetAvailableTimeZones()
    {
        // Return commonly used time zones
        return new[]
        {
            "Pacific Standard Time",
            "Eastern Standard Time",
            "Central European Standard Time",
```

100 % No issues found Ln: 36, Ch: 9 SPC CRLF Windows 1252

Diagnostic Tools

Diagnostics session: 10 seconds

10s

Add counter graphs by checking counters from counter options

Events

Process Memory (MB)

60 60

0 0

CPU (% of all processors)

100 100

0 0

Summary | Events | Counters | Memory Usage

Events

Show Events (0 of 0)

Application Insights Events (0 of 0)

.NET Counters

View Performance Counters

Memory Usage

Take Snapshot

CPU Usage

GitHub Copilot Chat

New Thread

Copilot Chat

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Use # for references, / for commands, and @ for extensions.

/help what can you do?

Reference

@

Ask Claude Sonnet 4.5

Solution Explorer Git Changes

Autos

Search (Ctrl+E) Search Depth: 3

Name Value Type

Autos Locals Watch 1

Developer PowerShell 7

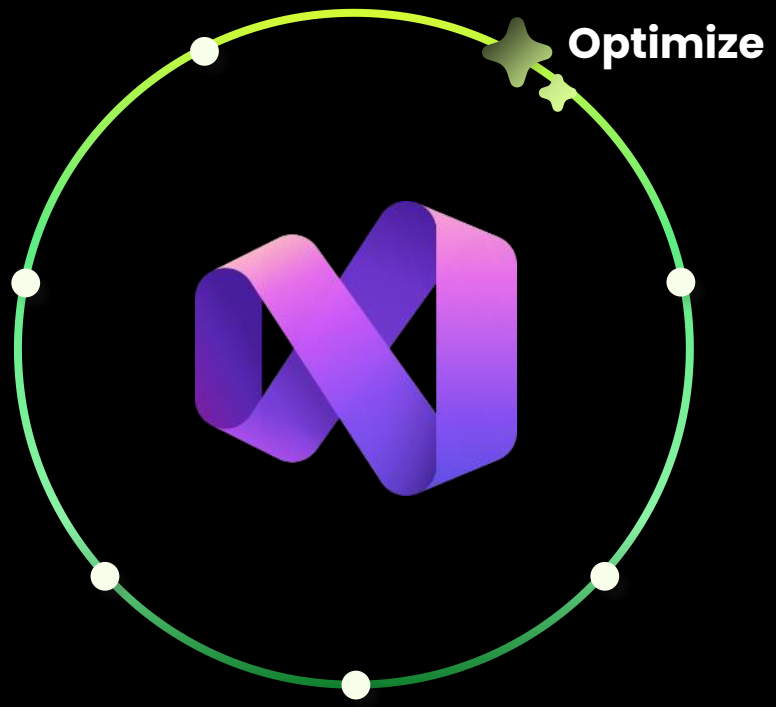
```
tibility purposes. If you want to re-enable it, run 'Import-Module PSReadLine'.

**********************************************************************
** Visual Studio 2026 Developer PowerShell v18.0.0-insiders
** Copyright (c) 2025 Microsoft Corporation
**********************************************************************
PS C:\Users\rachelkang\source\repos\TimeZoneHelperWebApp>
```

Developer Po... Developer Po... Call Stack Breakpoints Exception Set... Command W... Immediate W... Output

Ready 0 / 0 6 main TimeZoneHelperWebApp

1 inch of rain Thursday Search

Optimize



GitHub Copilot Chat

@Profiler Suggest how I can optimize my code

@Profiler Suggest how I can optimize my code

1 reference

Checkpoint created. Restore

Profiler

17 references

I'll help you optimize your code. Let me first understand the complete structure of your Todo application and then create benchmarks to identify performance bottlenecks.

4 tasks done

Now I'll create a plan to benchmark and optimize your code:

Plan Created Open

I'll begin by:

1. Creating a benchmark project to measure the performance of your TodoService and TodoRepository
2. Running CPU benchmarks to establish baseline performance
3. Analyzing the results to identify bottlenecks
4. Optimizing the code based on the benchmark data
5. Re-running benchmarks to verify improvements

Let me start by creating benchmarks for the key operations in your application.

Step 1: Create benchmark project for

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search TimeZoneHelperWebApp

Debug Any CPU TimeZoneHelperWebApp.Web https INSIDERS

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    - Dependencies
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    - wwwroot
    - Components
    - Services
    - appsettings.json
    - Program.cs
    - WeatherApiClient.cs

Solution Explorer | Git Changes

Output

Show output from: Build

GitHub Copilot Chat

New Thread

Copilot Chat

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/help what can you do?

+ Reference

Ask Copilot or use @workspace

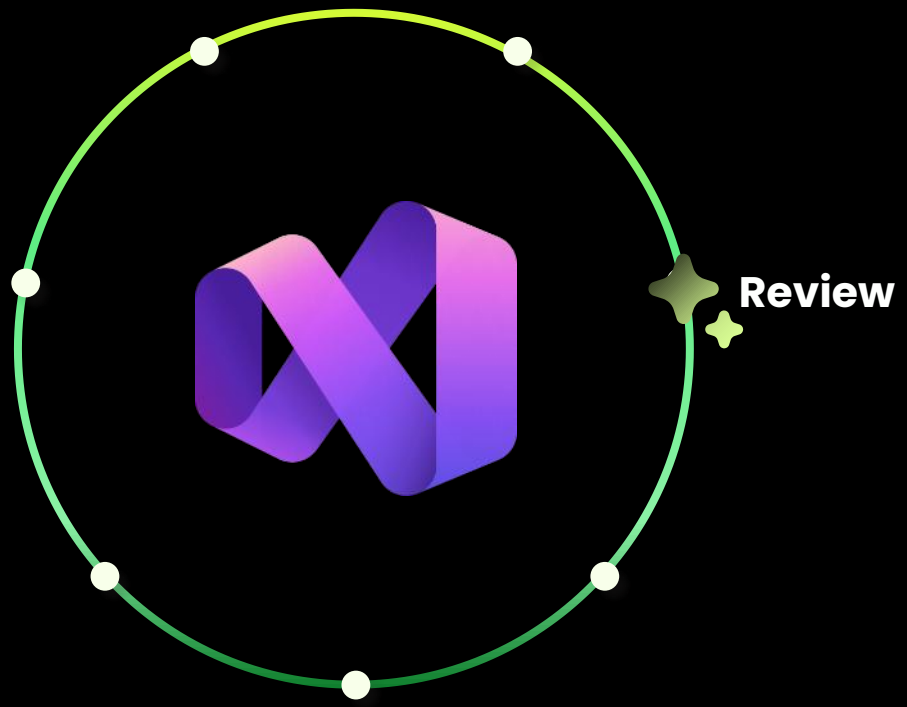
Agent Claude Sonnet 4.5

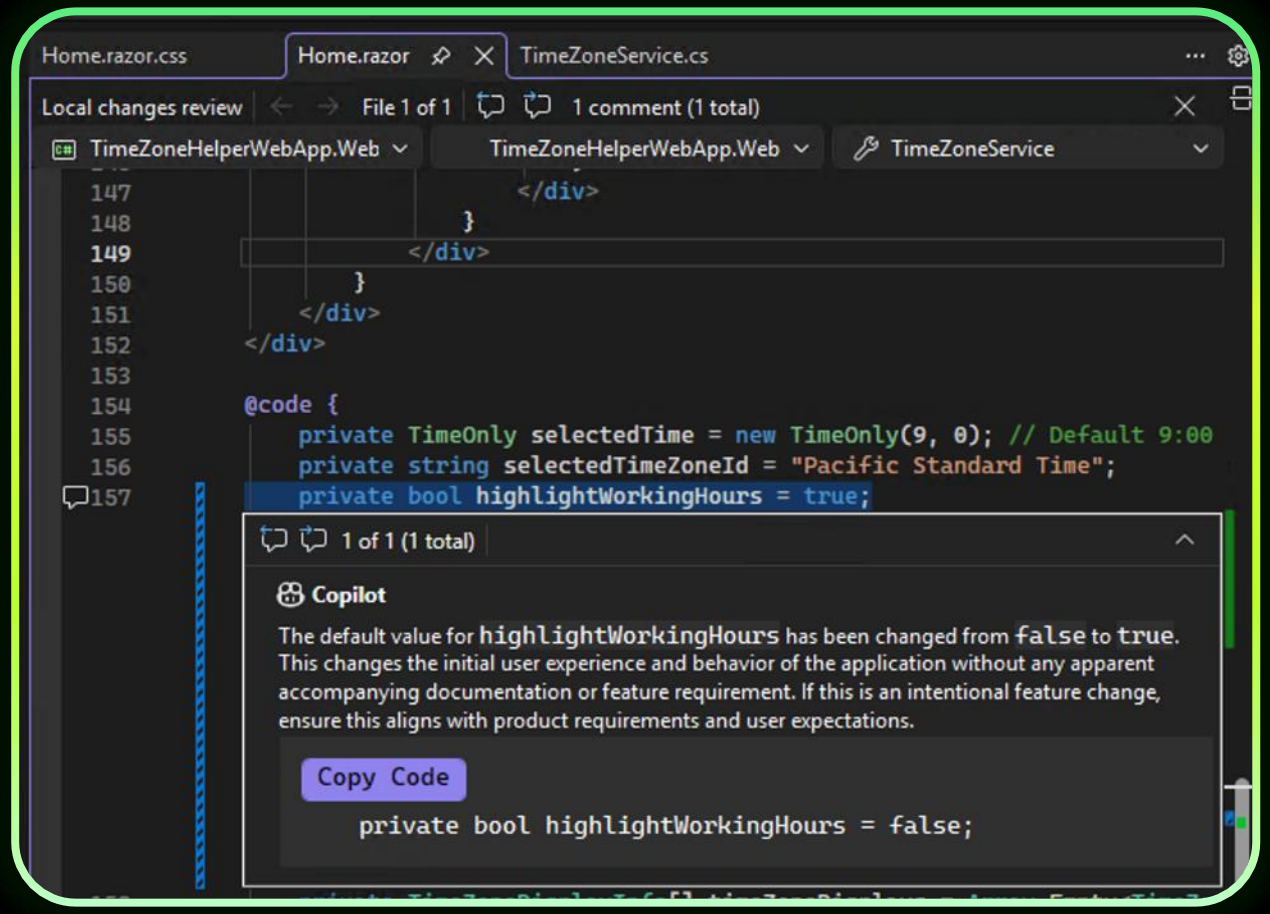
Diagnostic Tools

Ready 0 / 0 6 main TimeZoneHelperWebApp

Light rain At night Search

Review


Review


Home.razor.css
Home.razor
TimeZoneService.cs
Local changes review
File 1 of 1
1 comment (1 total)
TimeZoneHelperWebApp.Web
TimeZoneHelperWebApp.Web
TimeZoneService
147
148 }
149
150 }
151
152
153
154 @code {
155 private TimeOnly selectedTime = new TimeOnly(9, 0); // Default 9:00
156 private string selectedTimeZoneId = "Pacific Standard Time";
157 private bool highlightWorkingHours = true;
1 of 1 (1 total)
Copilot
The default value for highlightWorkingHours has been changed from false to true. This changes the initial user experience and behavior of the application without any apparent accompanying documentation or feature requirement. If this is an intentional feature change, ensure this aligns with product requirements and user expectations.
Copy Code
private bool highlightWorkingHours = false;

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search TimeZoneHelperWebApp

Debug | Any CPU | TimeZoneHelperWebApp.Web | https | INSIDERS

Git Changes - TimeZoneHelperWebApp

main

0 / 0 View all commits

Enter a message <Required>

Commit All | Amend

Related Items

Changes (2)

C:\Users\rachelkang\source\repos\Ti...

TimeZoneHelperWebApp.Web\Co...

Home.razor M

Home.razor.css M

Stashes

Home.razor | Home.razor.css | plan-c7e4620...25d55c6a7.md | Program.cs | Extensions.cs | TimeZoneService.cs

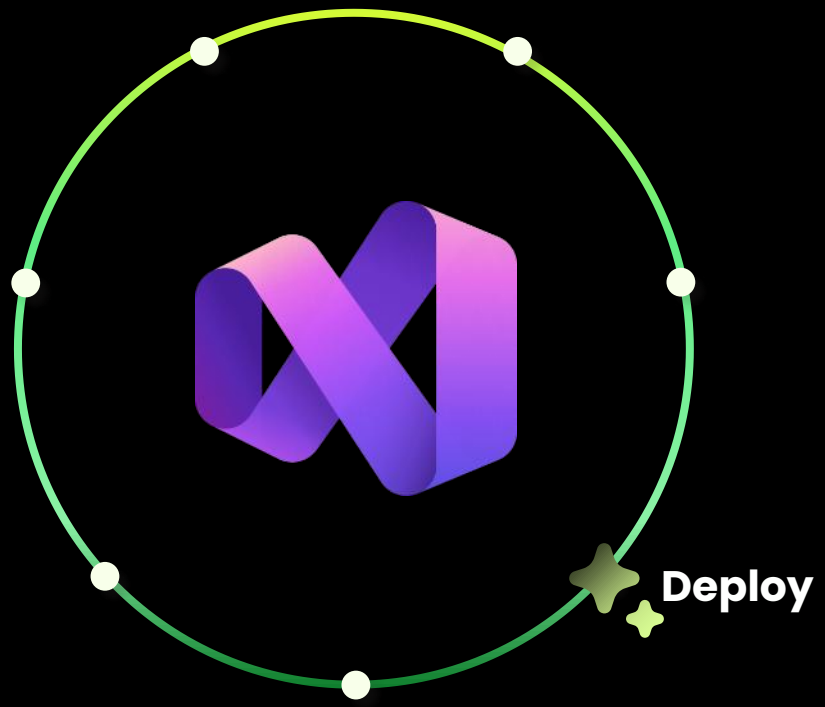
TimeZoneHelperWebApp.Web | TimeZoneHelperWebApp.Web.Components.Pages.H | TimeZoneService

```
86
87  @if (highlightWorkingHours && timeZoneDisplays.Length > 0)
88  {
89   <div class="working-hours-summary">
90   @{
91   var workingLocations = timeZoneDisplays.Where(tz => tz.IsWorkingHours).Select(tz => tz.DisplayName).ToList();
92    var offHoursLocations = timeZoneDisplays.Where(tz => !tz.IsWorkingHours).Select(tz => tz.DisplayName).ToList();
93        }
94
95  @if (workingLocations.Any() && offHoursLocations.Any())
96          {
97    <p>
98   <strong>Coworkers in @string.Join(", ", workingLocations)</strong> are still online.
99     It's currently outside working hours for coworkers in <strong>@string.Join(", ", offHoursLocations)</strong>.
100        </p>
101 }
102      else if (workingLocations.Any())
103        {
104       <p>
105 <strong>Coworkers in @string.Join(", ", workingLocations)</strong> are online.
106    </p>
107   }
108    else if (offHoursLocations.Any())
109        {
110       <p>
111            It's currently outside working hours for all locations.
112       </p>
113      }
114    </div>
115  }
116
117   <div class="timezone-grid">
118       @foreach (var tzInfo in timeZoneDisplays)
119       {
120 <div class="timezone-card @GetCardClass(tzInfo)">
121       <div class="timezone-header">
122           <h3>@tzInfo.DisplayName</h3>
123             <div class="timezone-header-actions">
124       <span class="timezone-offset">@tzInfo.OffsetDisplay</span>
```

100 % | No issues found | Ln: 98, Ch: 33 | SPC | CRLF | UTF-8 with BOM

```
******* ******* ******* 
```

Deploy



GitHub Copilot Chat

Deploy to Azure

Deploy to Azure

1 reference

GitHub Copilot

1 reference

I'll help you deploy this ASP.NET Core web application to Azure. Let me start by gathering some information about your project and then guide you through the deployment process.

Read 1 project

Ran verify_azure_publish_compatibility

Running Azure Best Practices

Run Azure Best Practices

Tool arguments:

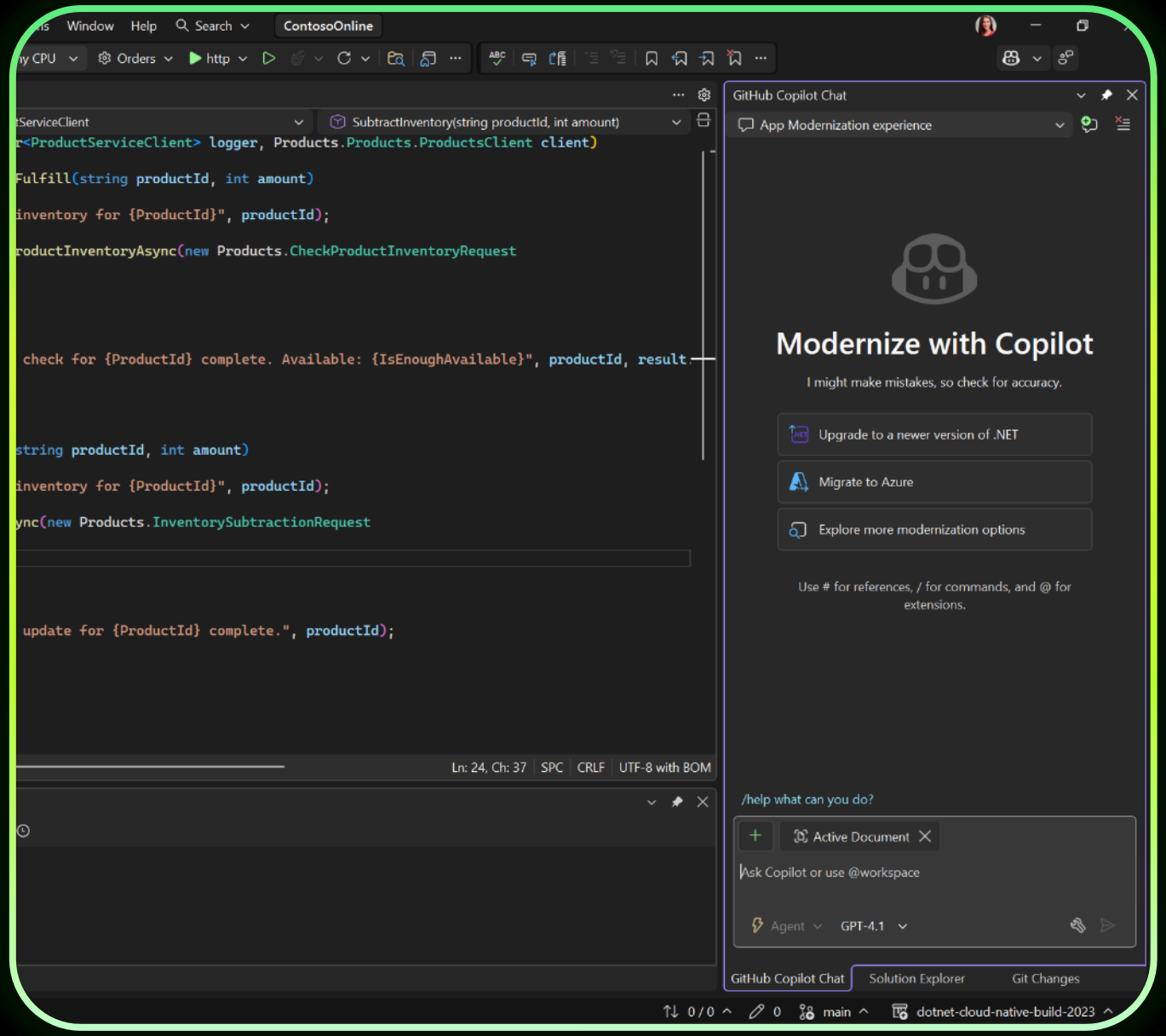
Name	Value
intent	Deploy ASP.NET Core web application to Azure

This tool is from 'Azure MCP Server' (MCP Server). Note that MCP servers or malicious conversation content may attempt to misuse Visual Studio through tools. Please carefully review any requested actions.

Confirm Deny

Modernize

Modernize



AI in Visual Studio 2026

Get Started with Copilot Free right in your favorite IDE

GitHub Copilot Dev Days

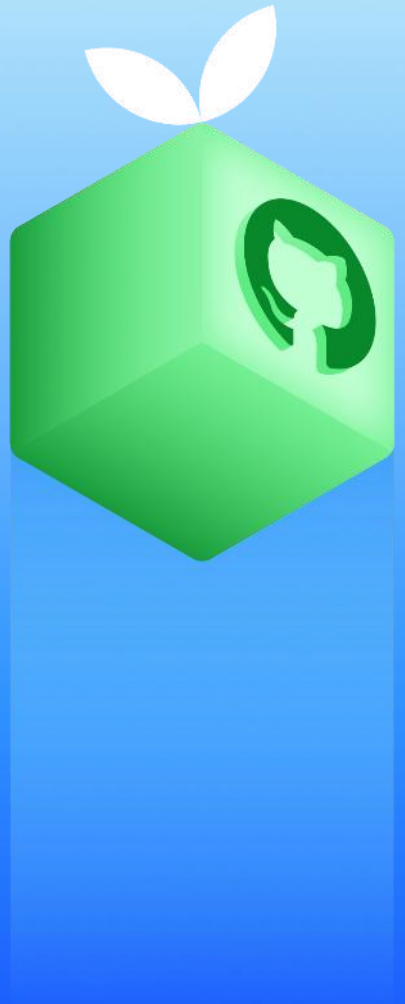
Demo

Visual Studio + GitHub Copilot

GitHub Copilot Dev Days Sweepstake

All completed surveys are eligible for our sweepstakes to win a GitHub Copilot Pro+ one-year subscription.

https://aka.ms/githubcopilotdevdays/sweepstakes



Thank you

gh.io/copilot